



Course Syllabus INTC 1343 H01A Application of Industrial Automatic Control

Instructor/Department Information

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Executive Director: Industrial & Technical Programs: Dr. Richard Davis

Vice President: Academic Affairs: Dr. Jenena Norris

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Course Details

Fall 2026

Session: Regular Academic Session

Location: TV2-119

Day(s): Tuesday Thursday

Times: 05:30 PM-06:20 PM

Lab Location: TV2-119

Lab Day(s): Tuesday Thursday

Lab Time(s): 06:25 PM-07:15 PM

Course Description

Automatic process control including measuring devices, analog and digital instrumentation, signal transmitters, recorders, alarms, controllers, control valves, and process and instrument diagrams. Includes connection and troubleshooting of loops.

Lecture Hrs = 2, Lab Hrs = 2

Credit Hours: 3

Student Learning Outcomes

1. Explain pneumatic and electronic controls; connect and troubleshoot control loops; Draw process and instrument diagrams, wiring diagrams, and block diagrams.

Prerequisites/Co-requisites

Prerequisite: INTC 1456 or DFTG 2408

Student/Office Hours

Students are encouraged to schedule a meeting with the course instructor for any questions or concerns regarding the course. In order to schedule a meeting, please contact the course instructor using the email or phone number listed above. Please allow 2 business days for a response. See instructor office hours below:

Student/Office Hours

Day	Hours
Monday	11:30am to 12:30pm
Tuesday	4:30pm to 5:30pm
Wednesday	11:30am to 12:30pm
Thursday	4:30pm to 5:30pm
Friday	

Required Textbook/Supplies Requirement

Title: Instrumentation and Process Control

ISBN: 9780826934468

Authors: Kirk, Weedon, Kirk

Publisher: American Technical Publishers

Edition: 7

required/optional: Required

Material Type: PAPERTEXTB

Title: Instrumentation and Process Control Workbook

ISBN: 9780826934475

Authors: Weedon

Publisher: American Technical Publishers

required/optional: Required

Material Type: PAPERWORKB

Title: KIT FOR INTC 1343

required/optional: Required
Material Type: SUPPLIES

myBooks Information

Non-dual-credit students are automatically enrolled in [myBooks](#), which provides digital and paper textbooks and other course materials. The cost of myBooks is \$33.00 per semester credit hour and applies to courses that have a required textbook or course kit. Students who wish to purchase their own books can opt out of myBooks by selecting the opt-out link in myLC Campus. If you need assistance, contact the Instructional Materials Manager at myBooks@lee.edu. For more information, please visit <https://www.lee.edu/mybooks/>.

Dual Credit students are not automatically opted in to myBooks. Please meet with your Lee College Dual Credit Advisor to learn more about the textbook process. You can find your dual credit advisor by visiting <https://www.lee.edu/dual/meet/>.

Technology

In our efforts to prepare students for a changing world, students may be expected to utilize computer technology in this course. Students may utilize available computer applications by logging into their One Login account located on the [Lee College Login webpage](#).

All emails from the college are sent to your myLC email account. Directions for forwarding your MyLC email to your personal email address are listed on the MyLC email page; however, emails from the college may include private educational records or other private information and the college is not responsible if this information is forwarded to an insecure personal email account.

Schedule/Outline

COURSE RUBRIC – COURSE SECTION: INTC 1343 H01A 2299 LECT/H01B 2300 LAB
COURSE NAME: Applications of Automatic Industrial Control
Term: FALL 2026
Instructor: Sajjad Khan

WEEK	DATE	CHAPTER	TOPIC(s)	ASSIGNMENTS	TEST/EXAMS

WEEK	DATES	CHAPTER	TOPICS	ASSIGNMENTS	TESTS/EXAMS
1	Aug 24 to 27	2	Fundamentals of Process Control		
2	Aug 31 to Sept 3	3	P & ID		
3	Sept 7 to 10	45	Safety devices	Due date for assignment 2 and 3 submission is Sept 8	Test 1 on Sept 8
4	Sept 14 to 17	46	Electrical Safety		
5	Sept 21 to 24	47	Safety Instrumented system		
6	Sept 28 to Oct 1	48	General Control Techniques		
7	Oct 5 to 8	48	General Control Techniques	Due date for assignment 45 to 48 submission is Oct 6	Lab Test 2 on Oct 6, on 603 simulator
8	Oct 12 to 15	49	Temperature Control		
9	Oct 19 to 22	49	Temperature Control		
10	Oct 26 to 29	50	Pressure Control		
11	Nov 2 to 5	50	Pressure Control		
12	Nov 9 to 12	50	Level Control		
13	Nov 16 to 19	50	Level Control		

WEEK	DATES	CHAPTER	TOPICS	ASSIGNMENTS	TESTS/EXAMS
14	Nov 23 to 26	51	Flow Control		
15	Nov 30 to Dec 3	51	Flow Control	Due date for assignments 49 to 51 submission is Dec 1	Test 3 on Dec 1
16	Dec 7 to 10				FINAL EXAM

****Schedule is subject to change at the discretion of the instructor or because of unforeseen circumstances****

Late Work Policy

Assignments must be submitted within one week of the assigned date.

Late submission of assignments will get a penalty of 10%.

Grading Components

Grading

Assignment Types	How many of each Assignment	Scores/points for each	Percentage of total grade per Assignment type
HOMEWORK	2	10	20
TESTS	2	15	30
LAB	1	20	20
ATTENDANCE	1	10	10
FINAL EXAM	1	20	20
	Total	Total Points	100%

*Faculty reserve the right to change a grading component and will provide written notice via email and announcements in Blackboard should a grading component substantively change.

Grading Scale

Grading	
Final Average in Percent	Letter Grade
89.50 - 100	A
79.50 - 89.49	B
69.50 - 79.49	C
59.50 - 69.49	D
< 59.50	F
Incomplete - no credit (See details below)	I

<http://catalog.lee.edu/content.php?catoid=3&navoid=70#grades>

Attendance Policy

<http://www.lee.edu/attendance-pol/>

Please note the Lee College attendance policy indicated in the link and notify me by phone or via email when you are not able to attend class. For the online attendance policy, in conjunction with the U.S. department of Education please follow the online attendance policy link above to review the policy.

Absences Due to Religious Holy Days

<http://www.lee.edu/absent-h-days/>

Students may be absent from classes for the observance of a religious holy day, provided they have given prior notice a week in advance and will be allowed to take an examination or complete an assignment scheduled for that day within a reasonable time as established by the instructor.

Withdrawal / Drop Policy

Students are responsible for dropping classes. State-mandated refund policies, as well as drop deadlines, appear in schedules, and on the [college website](#).

It is recommended students contact their advisor before they withdraw and drop a class.

Students who drop their courses after the official day of record and prior to the final day for Student Drops may be subject to the [Six Drop Rule](#).

Resignation (Complete Withdrawal):

Students may resign from all of their classes in any semester prior to the start of finals week. Students are required to sign a statement indicating they understand impacts of the resignation and are encouraged to visit with an advisor if they have concerns. A resignation results in a grade of W4 for all courses.

Incomplete

<http://www.lee.edu/i-grades/>

A grade of "I" indicates incomplete work resulting from illness or other unavoidable circumstances. To be eligible to receive an "I" students must have completed at least 75 percent of the work required for the course in question during the original term of enrollment. To receive an "I," a student must enter into a contract with the instructor of the course regarding the work that is to be completed and the grade the student will receive in the event that the work is not completed. The remaining work must be completed within one semester unless extenuating circumstances require a longer period for completion. Instructors have the right to submit any grade at any time to replace an "I" grade, with a grade of "F." Students who receive "I" grades should not re-enroll for the class unless they are terminating the incomplete agreement and wish to start over with a new section of the class. In this case, the student is urged to contact the original instructor to request release from the incomplete agreement.

After the next long semester has lapsed, if the "I" grade has not been changed to another grade by the instructor, the "I" grade will be replaced with an "F".

Incompletes (noted by the grade "I") are not issued in developmental courses.

Course-Specific Policies

If you are enrolled in a traditional face-to-face course, ignore the following.

If you are enrolled in a HYBRID or an ONLINE class, please read carefully the following:

HYBRID COURSES

This hybrid class requires you to complete half of the coursework on your own time using Blackboard. The other half of the course will be conducted through routine face-to-

face meetings. It is essential to stay on schedule, as hybrid classes can be more demanding than traditional face-to-face courses.

If your hybrid class includes a lab component, the labs will be completed during face-to-face sessions, while the lecture component will be conducted online.

If you have any questions about this course, please don't hesitate to reach out. Do not wait until you fall significantly behind.

For issues logging into Blackboard, contact the Lee College Help Desk at helpdesk@lee.edu or by phone at (281) 425-6952.

ONLINE COURSES

This online class requires you to complete all coursework independently using Blackboard. There will be no routine face-to-face meetings for this course.

It is essential to stay on schedule, as online classes can be more demanding than traditional face-to-face courses. Please note that online classes in Technical Studies do not include a lab component.

If you have any questions about this course, don't hesitate to reach out. Do not wait until you fall significantly behind.

For assistance with logging into Blackboard, contact the Lee College Help Desk at helpdesk@lee.edu or by phone at (281) 425-6952.

Academic Honesty Policy

<http://www.lee.edu/aca-honesty/>

Lee College expects and encourages all students to contribute to such an environment by observing the principles of academic honesty outlined in the College's Academic Honesty Code, which is detailed in Chapter Three of the Lee College Catalog under Academic Honesty. It is the student's responsibility to understand and comply with this code.

AI Policy

AI Encouraged with Attribution, Verification, and Reflection: Using AI to find information, brainstorm ideas, and test arguments is encouraged and expected. Copying and/or paraphrasing from an AI assistant is permitted as long as copied material is quoted or in a different colored font with a citation, and paraphrased material has a citation. Citations must include verification. To ensure a proper citation, follow these guidelines:

- The citation should indicate that the material was copied or paraphrased from an AI assistant.
- If the AI provided facts, sources, and/or quotes, the citation should identify the facts, sources, and/or quotes and confirm that they were independently verified and that they are correct and/or real.

In addition to copying and paraphrasing, you can present ideas and arguments provided by the AI assistant just as you can present ideas and arguments provided by other people; however, like other people, AI assistants can be biased even when the facts, quotes, and sources are real. Therefore, when using ideas from an AI assistant, you are required to reflect on the ideas and consider biases that may be present.

Reason: AI has become part of the network of people and technology that we use to gather information and ideas. Using AI is encouraged because it is important to understand how to exercise control over AI and engage thoughtfully with it. This includes verifying that facts, quotes, and sources are real and that ideas and themes presented by the AI assistant are susceptible to bias.

General AI Principles:

- AI should help you think. Not think for you.
- You are responsible for the accuracy of facts, quotes, sources, etc., provided by your AI assistant.
- Do not enter information into AI tools that you would not post on social media.
- You must disclose when you use generative AI in your work.
- The AI policy in this course may be different from the policy in other courses.

DEI Policy

Effective January 1, 2024, Senate Bill 17 (SB 17) prohibits community colleges from engaging in certain diversity, equity, and inclusion (DEI) initiatives, with some exceptions. One of the exceptions is that SB 17's prohibitions do not apply to academic courses; scholarly research or creative works by students, faculty, or research personnel or the dissemination of the research or works; or data collection. This means that faculty can

include DEI topics in class discussion, readings, research projects, homework assignments, and other aspects of instruction, and students and faculty can exhibit and present their work without restrictions.

Non-Discrimination:

The College District prohibits discrimination, including harassment, against any student on the basis of an individual's Protected Characteristics including race, color, national origin, disability, religion, age, limited English proficiency or English learner status, veteran status, genetic information, sex (including pregnancy, parental status, sex stereotyping or treating people differently because they do not conform to sex-role expectations, sexual orientation, gender identity or gender expression) or any other basis prohibited by law. Retaliation against anyone involved in the complaint process is a violation of College District policy and is prohibited.

Complaints/Grievances

<http://catalog.lee.edu/content.php?catoid=27&navoid=750&hl=appeals#student-complaints>

Students are encouraged to contact their instructor for assistance regarding any aspect of the course. For issues that are not resolved after consultation with the instructor, students may contact the Division Chair. If an issue is not resolved after meeting with the instructor and division chair, the student can submit a formal complaint online by following the procedures in the link above.

Student Support and Services

<https://www.lee.edu/student-services/>

Lee College is proud to offer a variety of student services to provide a variety of supports within our division to assist students navigate their time at the college. You can find more information about student services at the link above.

Timely Care

[TimelyCare](#) is available to current credit students for free, providing virtual care to include mental health support, medical visits, and general well-being information. Download the app today by going to www.timelycare.com/lee

Access Center for Students with Disabilities

The Lee College [Access Center for Students with Disabilities](#) is located in Rundell Hall Office 101 and is the campus Disability Services unit that works with students who have disabilities to provide and/or arrange reasonable accommodations for students in their courses. Students who have, or think they may have, a disability, are invited to contact the Access Center for a confidential discussion at (832)-556-4069 or at kvillanueva@lee.edu.

[Link to the Americans with Disabilities Act \(ADA\) Website](#)

kNOw More (Title IX)

<https://www.lee.edu/titleix/know-more-campaign/>

Lee College is committed to providing all with a learning and work environment that is free from sexual harassment and assault. If you or someone you know has been harassed or assaulted, you can find the appropriate resources, both on and off campus at the [Title IX Available Resources Page](#). Confidential support can be found through The Bridge Over Troubled Waters. The Bridge can be contacted at 281-420-5600 (Baytown Office) or through their 24-hour hotline at 713-473-2801. Here is the link for the U.S. Department of Education – [Title IX – Sex Discrimination](#).

Lee College faculty are “responsible employees”, which means, as a faculty member I am required to report incidents of sexual misconduct and thus cannot guarantee confidentiality. I must provide the Title IX Coordinator with relevant details such as the names of those involved in the incident. If you want to make a formal report, you can contact Jose Martinez, Title IX Coordinator, at josmartinez@lee.edu or 281-425-6546 or email TitleIXSupport@lee.edu.